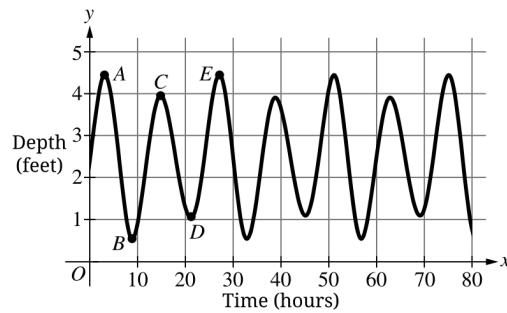


AP Precalculus Classwork 3.1–3.4

Name: _____

Date: _____

Section 3.1 — Period Phenomena



A graph of the depth of water at a pier in the ocean is given, along with five labeled points A , B , C , D , and E in the xy -plane. For the time periods near these data points, a periodic relationship between depth of water, in feet, and time, in hours, can be modeled using one cycle of the periodic relationship. Based on the graph, which of the following is true?

☐ A

The time interval between points A and B gives the period.

☐ A

☐ B

The time interval between points A and C gives the period.

☐ B

☐ C

The time interval between points A and D gives the period.

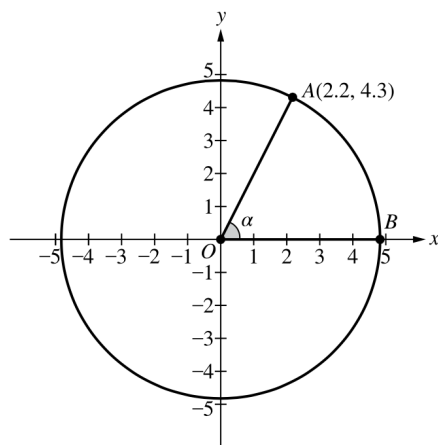
☐ C

☐ D

The time interval between points A and E gives the period.

☐ D

Section 3.2 — Sine, Cosine, & Tangent



The figure gives a circle of radius 4.8 in the xy -plane with center at the origin, an angle α in standard position, and three labeled points. Which of the following is the value of $\sin \alpha$?

(A)

$$\frac{2.2}{4.8}$$

☐ (A)

(B)

$$\frac{4.3}{4.8}$$

☐ (B)

(C)

$$2.2$$

☐ (C)

(D)

$$4.3$$

☐ (D)

In the xy -plane, angle BAC is an angle in standard position with terminal ray AC , which intersects the unit circle at the point with coordinates $(0.4, -0.9)$. Which of the following descriptions is correct?

(A)

The tangent of angle BAC is $-\frac{4}{9}$, and the slope of ray AC is $-\frac{4}{9}$.

☐ (A)

(B)

The tangent of angle BAC is $-\frac{4}{9}$, and the slope of ray AC is $-\frac{9}{4}$.

☐ (B)

(C)

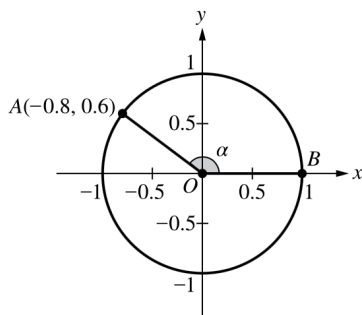
The tangent of angle BAC is $-\frac{9}{4}$, and the slope of ray AC is $-\frac{9}{4}$.

☐ (C)

(D)

The tangent of angle BAC is $-\frac{9}{4}$, and the slope of ray AC is $\frac{4}{9}$.

☐ (D)



The figure shows a unit circle in the xy -plane, an angle α in standard position, and three labeled points.

Which of the following is the value of $\cos \alpha$?

☐ (A) -0.8

☐ (A)

☐ (B) -0.6

☐ (B)

☐ (C) 0.6

☐ (C)

☐ (D) 0.8

☐ (D)

Section 3.3 — Sine and Cosine Function Values

In the xy -plane, an angle in standard position measures $\frac{5\pi}{6}$ radians. A circle centered at the origin has radius 4. What are the coordinates of the point of intersection of the terminal ray of the angle and the circle?

☐ (A) $(-2\sqrt{3}, 2)$

☐ (A)

☐ (B) $(-\frac{\sqrt{3}}{2}, \frac{1}{2})$

☐ (B)

☐ (C) $(\frac{\sqrt{3}}{2}, \frac{1}{2})$

☐ (C)

☐ (D) $(2\sqrt{3}, 2)$

☐ (D)

In the xy -plane, the terminal ray of angle θ in standard position intersects a circle of radius r at the point $(10, -10\sqrt{3})$. What are the values of θ and r ?

☐ (A) $\theta = \frac{5\pi}{3}$ and $r = 10$



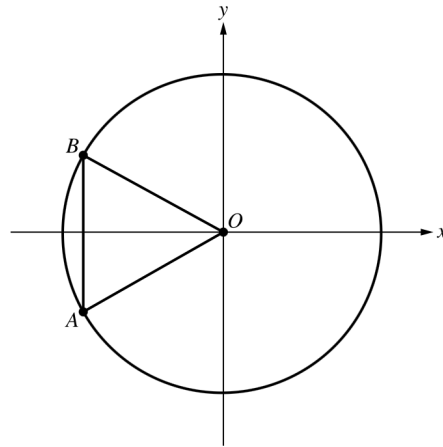
☐ (B) $\theta = \frac{5\pi}{3}$ and $r = 20$



☐ (C) $\theta = \frac{11\pi}{6}$ and $r = 10$



☐ (D) $\theta = \frac{11\pi}{6}$ and $r = 20$



The figure shows equilateral triangle ABO with sides of length 10 in the xy -plane. Segment AB is perpendicular to the x -axis. The terminal ray of an angle θ (not shown) in standard position passes through the point B . What is the value of $10 \sin \theta$?

☐ (A) $-5\sqrt{3}$



☐ (B) $-\frac{\sqrt{3}}{2}$



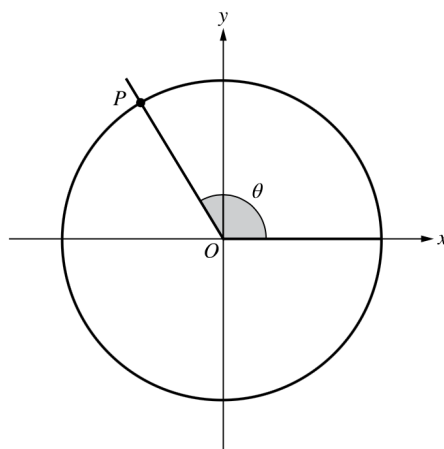
☐ (C) $\frac{1}{2}$



☐ (D) 5



Section 3.4 — Sine and Cosine Function Graphs



The figure shows the terminal ray of angle θ , in standard position, intersecting the unit circle at point P in the xy -plane. The function g is given by $g(z) = \cos z$. For the angle ω (not shown), $\theta < \omega < \pi$. Which of the following is true?

☐ (A) $g(\omega) < g(\theta)$



☐ (B) $g(\omega) > g(\theta)$



☐ (C) $g(\omega) = g(\theta)$



☐ (D) Depending on the value of ω , sometimes $g(\omega) < g(\theta)$ and sometimes $g(\omega) > g(\theta)$.



The function g is given by $g(\theta) = \cos \theta$. Which of the following describes g on the interval from $\theta = \frac{3\pi}{2}$ to $\theta = 2\pi$?

☐ (A) g is decreasing, and the graph of g is concave down.



☐ (B) g is decreasing, and the graph of g is concave up.



☐ (C) g is increasing, and the graph of g is concave down.



☐ (D) g is increasing, and the graph of g is concave up.



The function f is given by $f(\theta) = \sin \theta$. Which of the following describes f on the interval from $\theta = \pi$ to $\theta = \frac{3\pi}{2}$?

☐ A f is decreasing, and the graph of f is concave down.

☐ A

☐ B f is decreasing, and the graph of f is concave up.

☐ B

☐ C f is increasing, and the graph of f is concave down.

☐ C

☐ D f is increasing, and the graph of f is concave up.

☐ D